LAB 7: Collaboration Diagram

Smart Waste Collection Management System

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Prepared by: Tanishka Singh - Leader (Product Manager)

1. Objective

To draw UML collaboration (communication) diagrams for the main scenarios of the Smart Waste Collection Management System. A collaboration diagram shows the objects, the links between them and the numbered messages exchanged, so it highlights how objects are organised rather than only the time order. The scenarios are taken from SRS v1.0.

2. Collaboration diagrams included

Figure	Scenario	SRS reference
Diagram 1 (page 2)	Sensor measures and transmits the fill-level, server stores it, checks the threshold and sends push alerts to the administrator and driver.	4.1, 4.2 (REQ-BM-1 to 3, REQ-AL-1 to 3)
Diagram 2 (page 3)	Administrator requests a route, server computes it using the mapping API, admin may adjust it, route is sent to the driver who collects bins.	4.3, 5.5 (REQ-RO-1 to 3)
Diagram 3 (page 4)	Citizen submits a complaint and gets a tracking ID; it is marked Resolved when the bin is collected; citizen checks the status.	4.5, 5.5 (REQ-CF-1 to 3)

3. Objects

Object	Type	Role in the collaboration
:SensorNode	Object	Measures distance, calculates fill percentage and transmits it.
:BackendServer	Object	Stores readings, checks the threshold, raises alerts, computes routes and resolves complaints.
:Database	Object	Stores readings, alerts, routes, bins and complaints.
:PushService (FCM)	Object	Delivers alerts to the admin and driver apps.
:MappingAPI (OSM)	Object	Provides distance data for route optimization.
:MobileApp / WebForm	Object	Interface used by the citizen to submit and track complaints.
:Administrator, :Driver, :Citizen	Actor	Human users of the dashboard and mobile app.

4. Notation used

Symbol	Meaning
Rectangle with underlined :Name	Object (green = system object, blue = actor)
Solid line between two objects	Link - the objects communicate with each other
Small arrow beside a link, e.g. 3: transmitData()	Message with its sequence number and direction
Dotted numbers such as 3.1, 3.2	Messages triggered while handling message 3 (nesting)
[condition] before a message	Guard - the message is sent only if the condition is true
Loop on one object	Self message, sent from the object to itself

Collaboration Diagram 1 - Bin Fill-Level Monitoring and Threshold Alert

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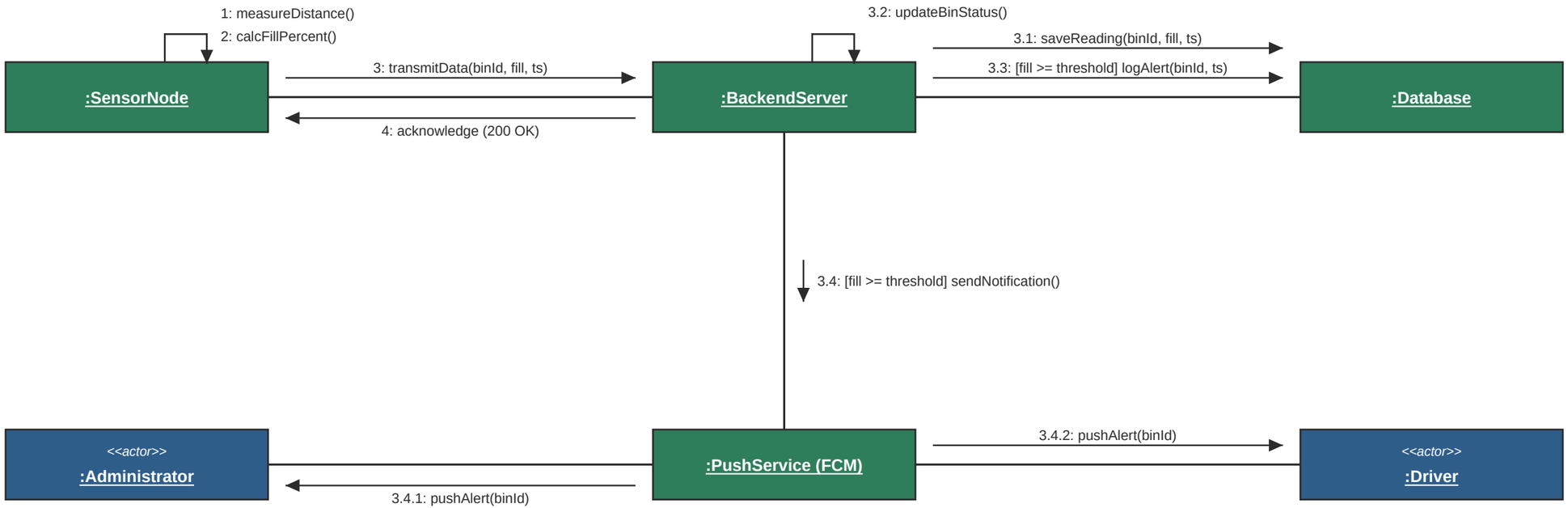

Numbers show the order of messages; dotted numbers (3.1, 3.2 ...) show messages triggered while handling message 3.

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Collaboration Diagram 2 - Route Optimization and Bin Collection

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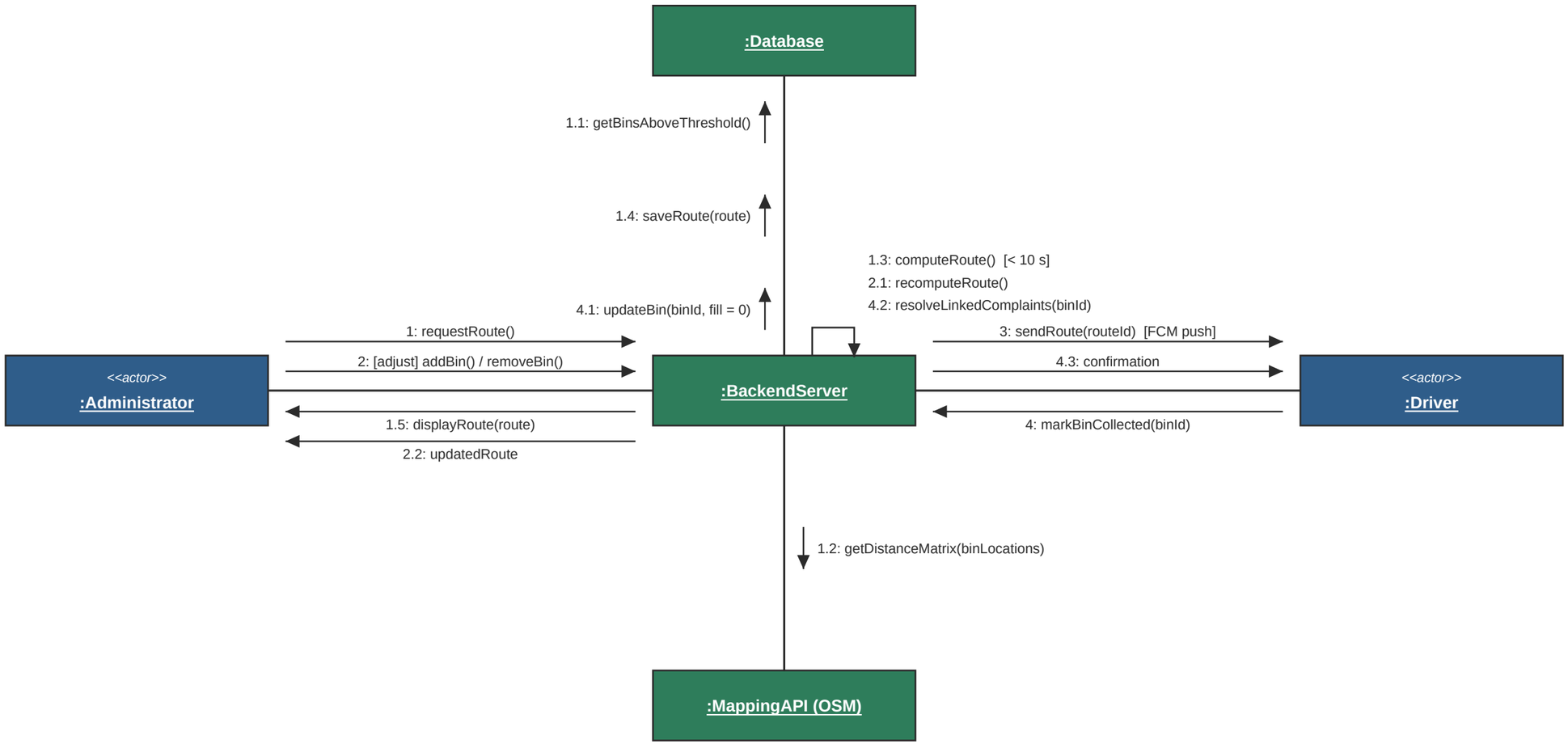

Numbers show the order of messages; dotted numbers (3.1, 3.2 ...) show messages triggered while handling message 3.

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Collaboration Diagram 3 - Citizen Complaint Handling

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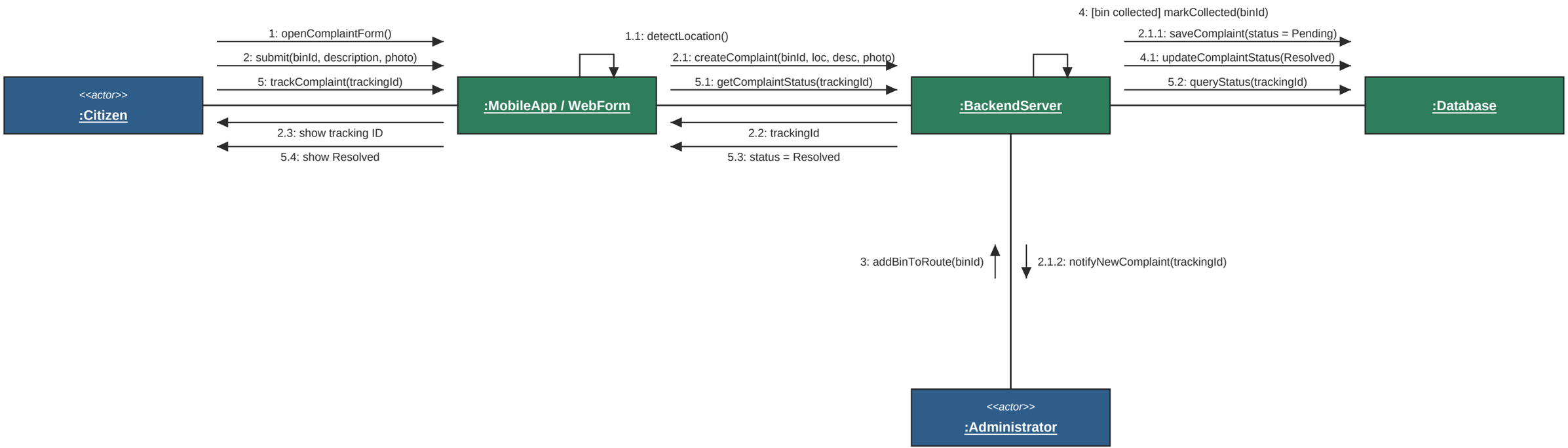

Numbers show the order of messages; dotted numbers (3.1, 3.2 ...) show messages triggered while handling message 3.

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